

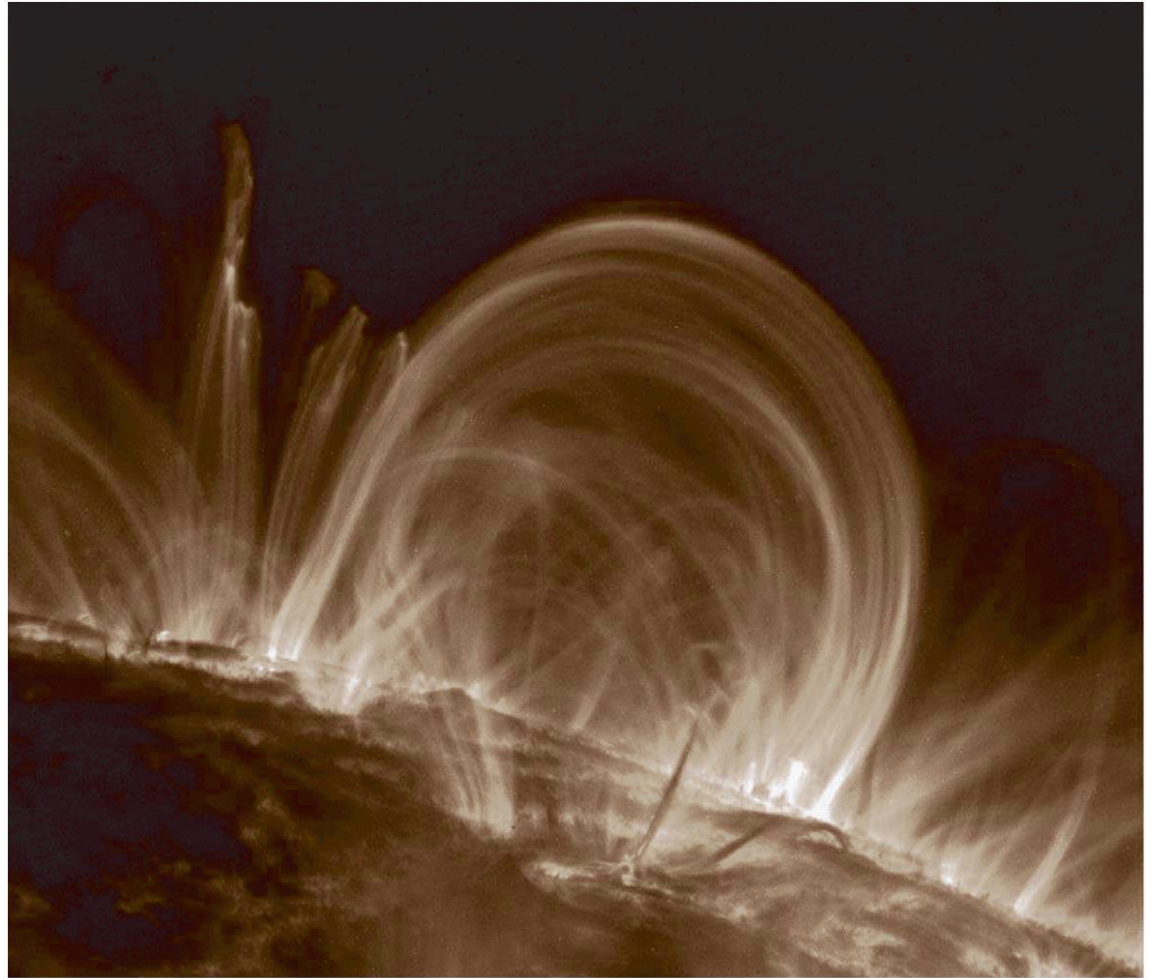
Ch 28 Magnetic field

This ultraviolet image shows delicate loops of million-kelvin ionized gas in the Sun's atmosphere (coronal mass ejection).

What force shapes the gas into such intricate structures, and why don't we see similar things in Earth's atmosphere?

Magnetic force.

Atmosphere on earth not ionized.



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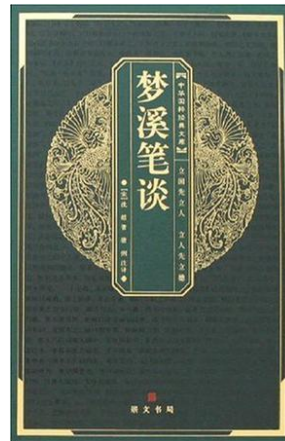
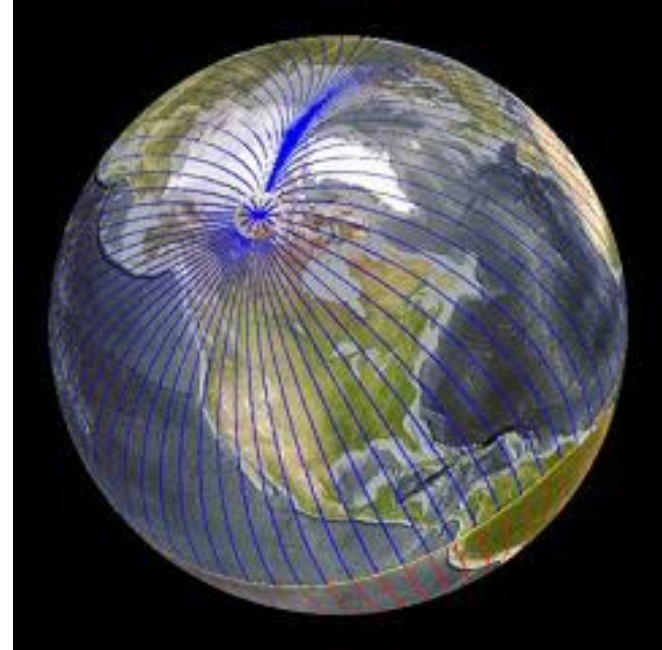
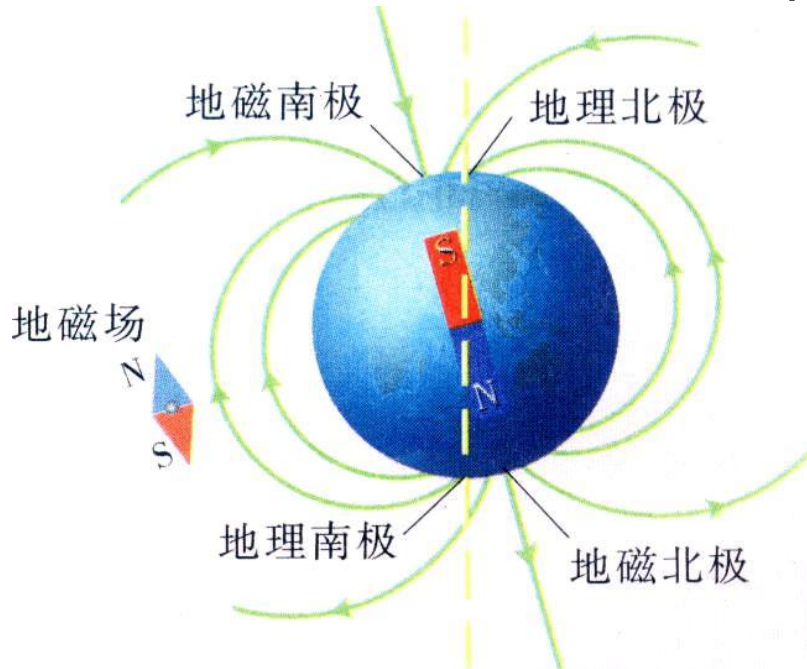
<https://haokan.baidu.com/v?vid=10360491245798342044&pd=bjh&fr=bjhauthor&type=video>

Today's topics (Chapter 28)

- Magnets
- Lorentz Force and Its Applications
- Hall Effect and Magnetic Force on a Current
- Magnetic Dipole

Earth's Magnetic Field

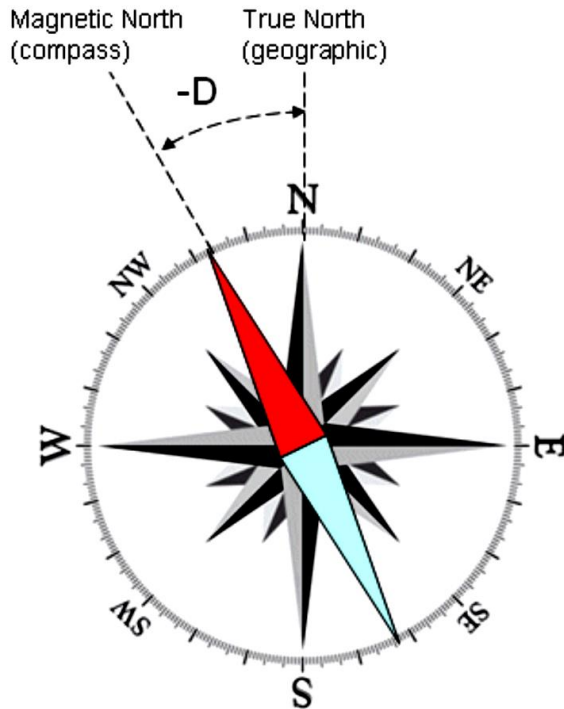
Migration of the magnetic *south* pole



方家以磁石磨针锋，
则能指南，然常微
偏东，不全南也。

How to use a compass correctly :

Negative Magnetic Declination: Magnetic North is WEST from True North:



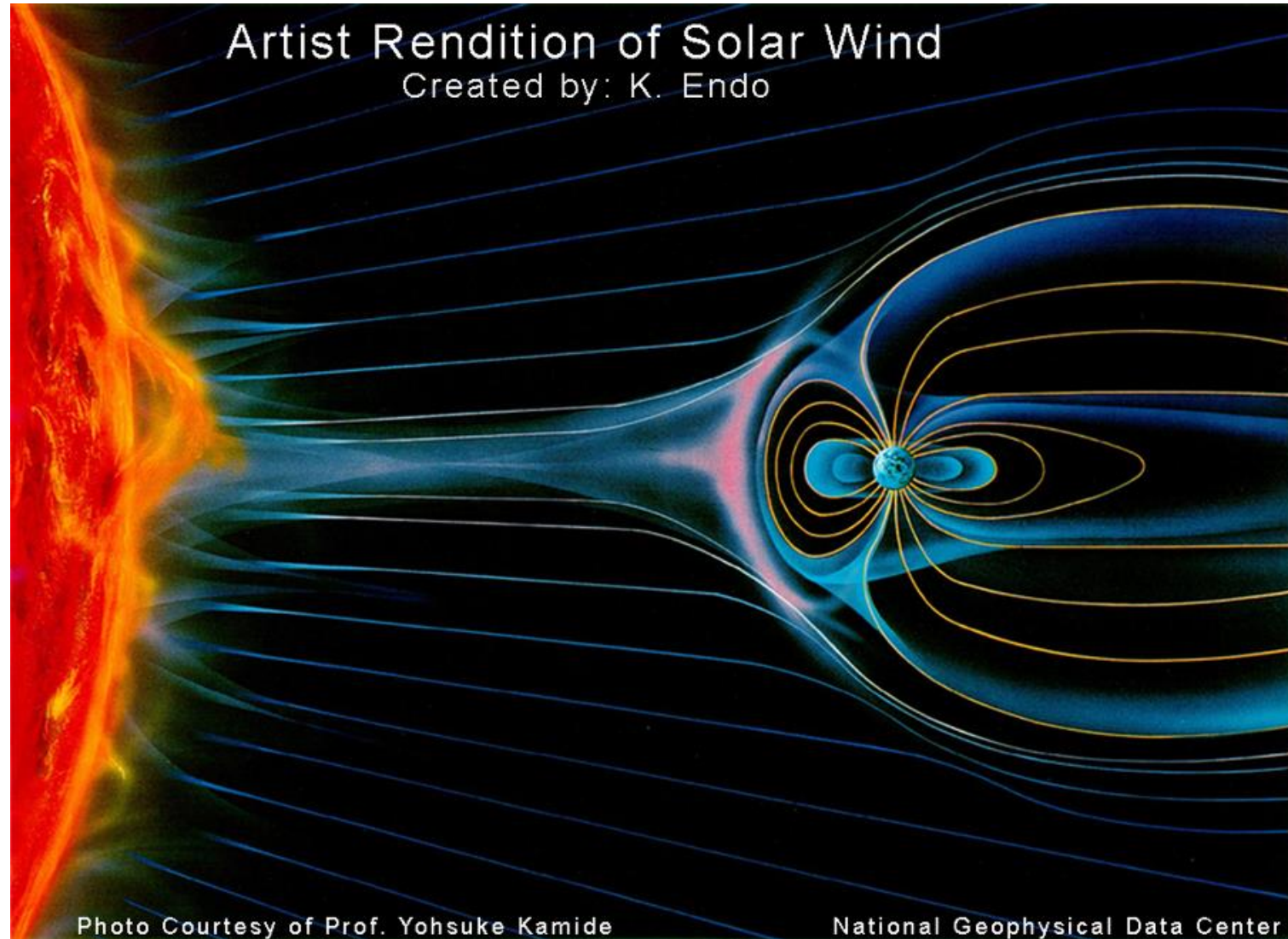
Magnetic declination at
Shenzhen : $2^{\circ}26''$ West



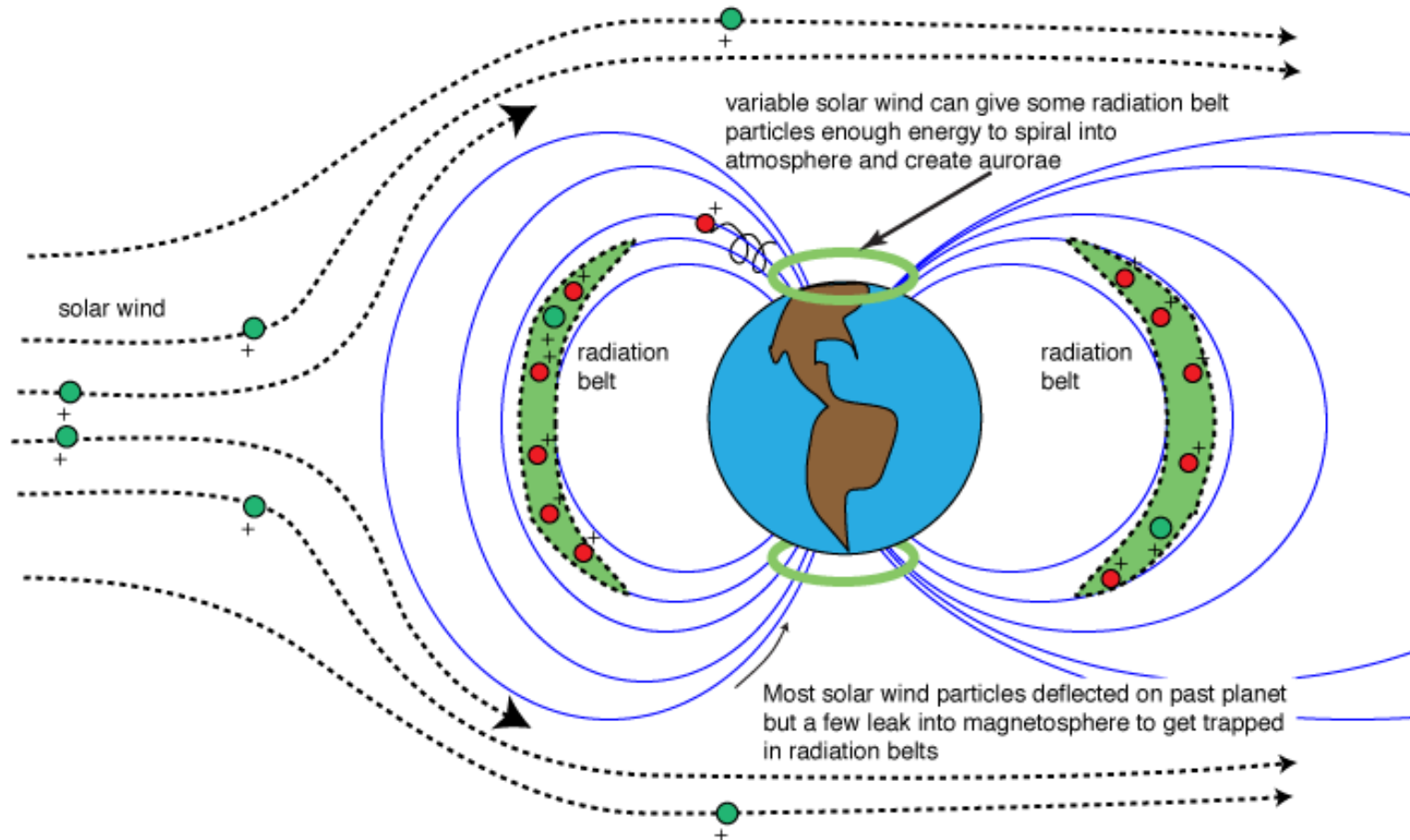
Aurora



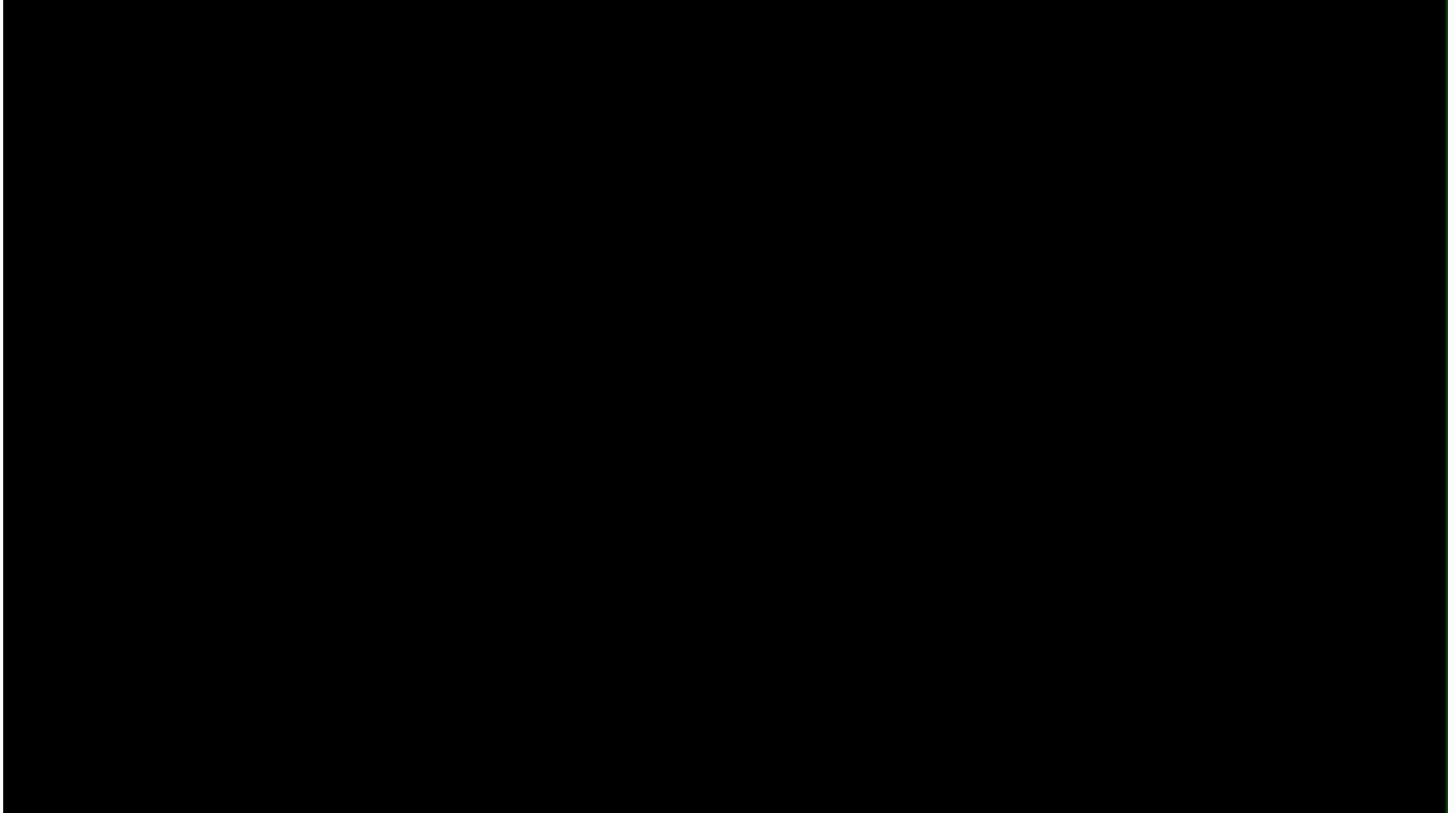
Solar wind and the aurora



Solar wind and the aurora



Van Allen radiation belt & aurora



Magnetic storage

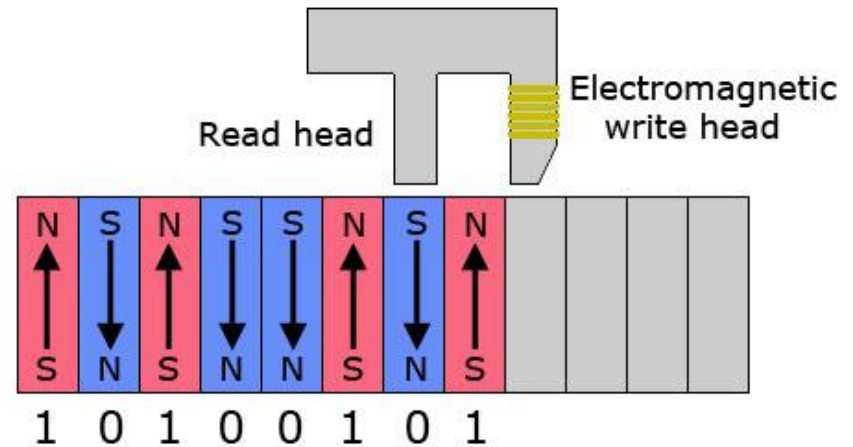


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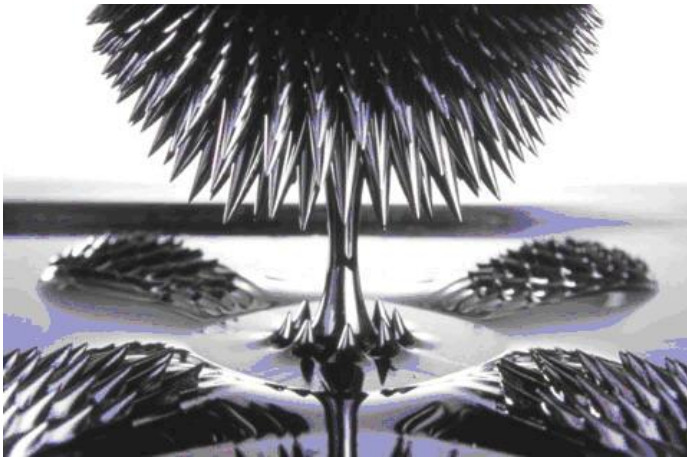


Hard drive read/write head



<http://www.computerhope.com>

Magnetic fluid



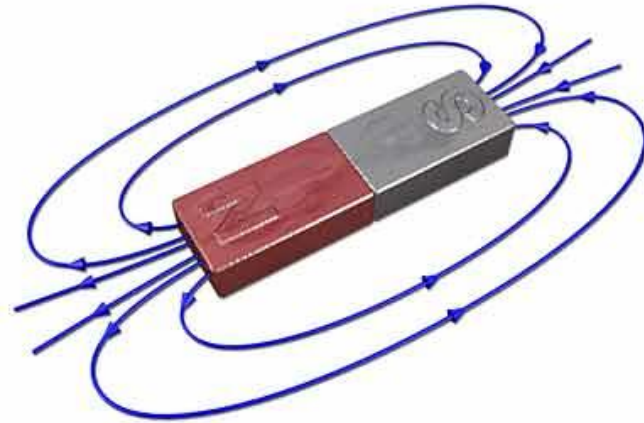
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What creates a magnetic field?

An electric field can be produced by a charge – an “electric monopole”. However, since there’s still no evidence for the *magnetic monopole* to exist, the magnetic field $\mathbf{B}$ is generally produced by the following methods.



An electromagnet



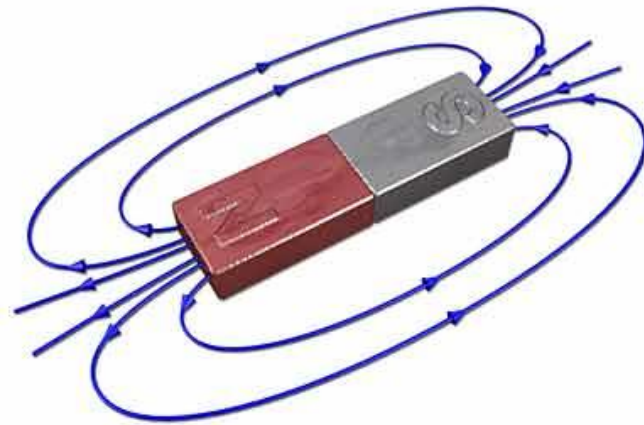
A permanent magnet

What creates a magnetic field?

1. **Moving charges (currents).** Moving charges or current can produce magnetic fields to make an *electromagnet*.
2. **Nuclear or electron spins.** Subatomic particles often associate with intrinsic magnetic fields called “spin”. Many spins align to give a macroscopic *permanent magnet*.
3. **Changing electric fields (electromagnetic waves).**



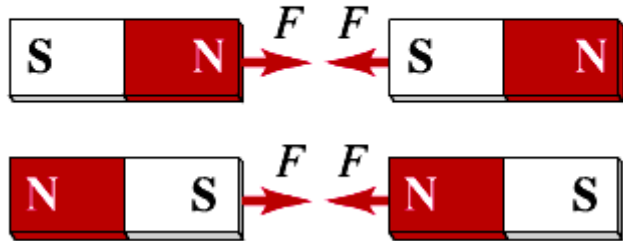
An electromagnet



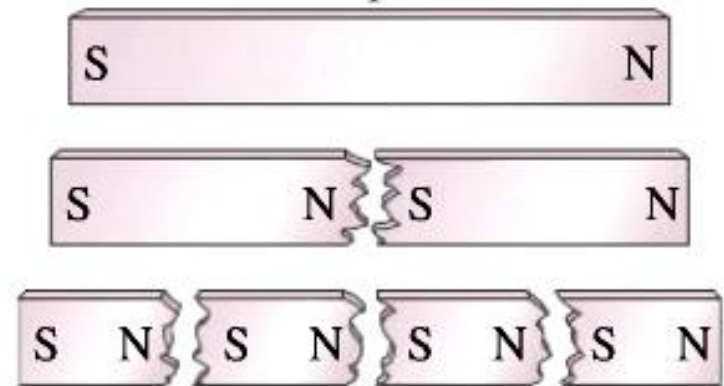
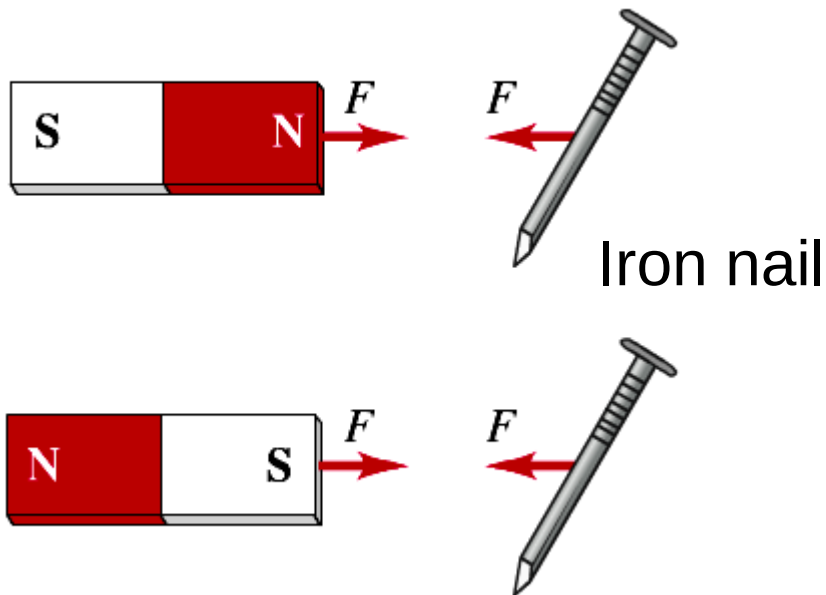
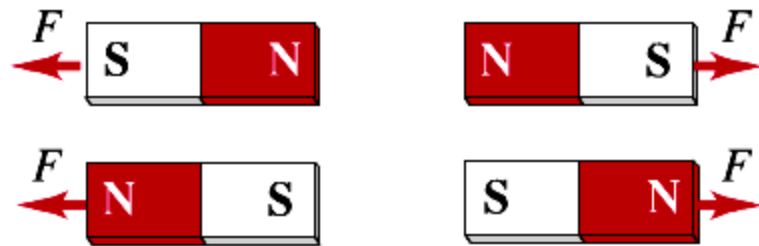
A permanent magnet

Magnets and magnetic forces

Opposite poles attract.



Like poles repel.



no magnetic monopole
observed so far

Electric and magnetic interactions

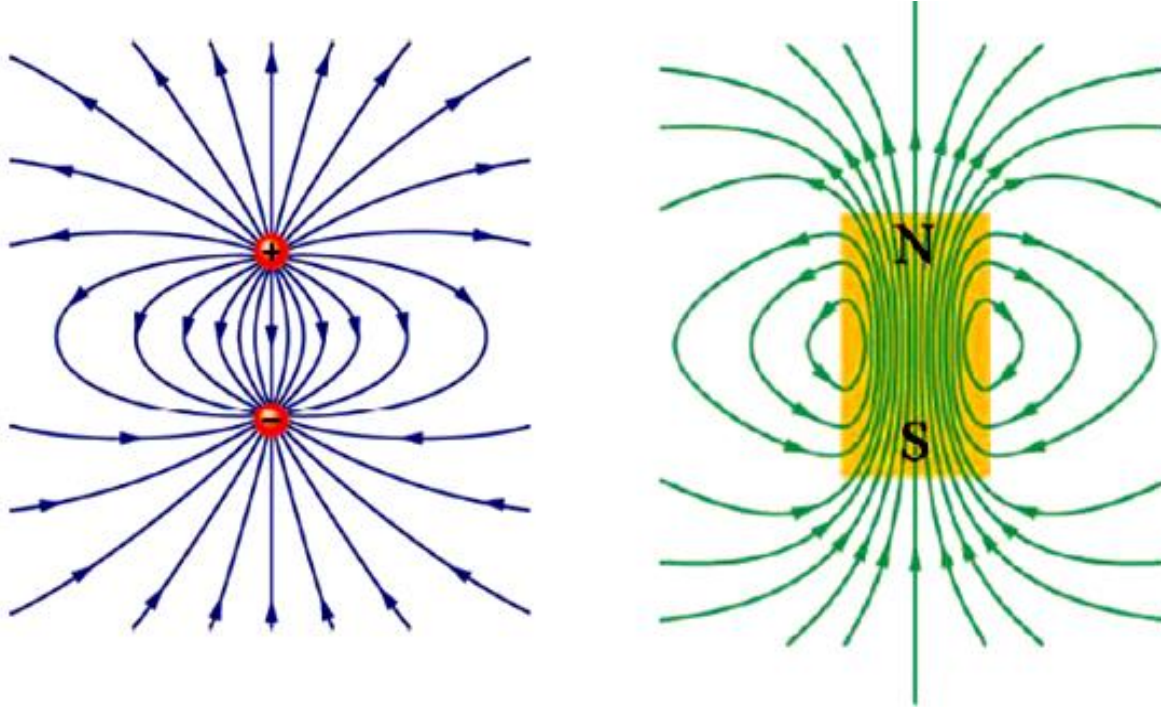
Electric interactions :

1. A distribution of electric charge at rest creates an electric field $\vec{E}$ in the surrounding space.
2. The electric field exerts a force $\vec{F} = q\vec{E}$ on any other charge q that is present in the field.

Magnetic interactions :

1. A moving charge or a current creates a **magnetic field** in the surrounding space (in addition to its *electric* field).
2. The magnetic field exerts a force $\vec{F}$ on any other moving charge or current that is present in the field.

Electric field vs. Magnetic field



- 1) Magnetic field lines have no ends → they form closed loops **Why?**
- 2) Electric force $\parallel$ Electric field lines,
Magnetic force $\perp$ Magnetic field lines (explained soon)
- 3) There is isolated charge but there is no isolated magnetic pole
- 4) Both are vector fields

Gauss' Law for magnetic fields

Magnetic Flux

Similar to the electric flux, the magnetic flux is defined as the total amount of magnetic field lines that goes through a surface A .

$$d\Phi_B = \mathbf{B} \cdot d\mathbf{A}; \quad \Phi_B = \int_A \mathbf{B} \cdot d\mathbf{A}$$

Unit. The SI unit of Φ_B is the weber (Wb).

$$1 \text{ weber} = 1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2.$$

Gauss' Law of the Magnetic Field

There's neither source nor sink in a magnetic field. **The net magnetic flux through a closed surface is always zero.**

$$\oint \mathbf{B} \cdot d\mathbf{A} = 0$$

Why?

The Second Equation of the Maxwell's Equations

Differential Form

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \end{array} \right.$$

Integral Form

$$\left\{ \begin{array}{l} \oint_{\mathbf{A}=\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0} \\ \oint_{\mathbf{A}=\partial V} \mathbf{B} \cdot d\mathbf{A} = 0 \quad \leftarrow \text{Gauss' Law of the B-field} \\ \oint_{\mathbf{l}=\partial \Sigma} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_{\Sigma} \mathbf{B} \cdot d\mathbf{\Sigma} \\ \oint_{\mathbf{l}=\partial \Sigma} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(\int_{\Sigma} \mathbf{J} \cdot d\mathbf{\Sigma} + \epsilon_0 \frac{d}{dt} \int_{\Sigma} \mathbf{E} \cdot d\mathbf{\Sigma} \right) \end{array} \right.$$

Gauss' law of the magnetic field tells us that there's neither source nor sink in a magnetic field. All magnetic field lines form close contours. Without the existence of magnetic monopoles, the Maxwell's equation is *not* symmetric between the $\mathbf{E}$ and $\mathbf{B}$ fields.

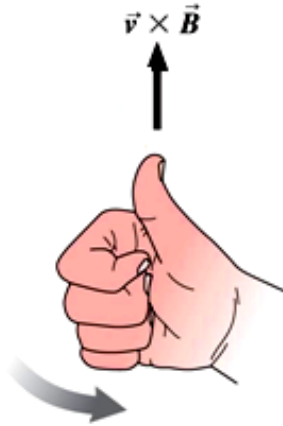
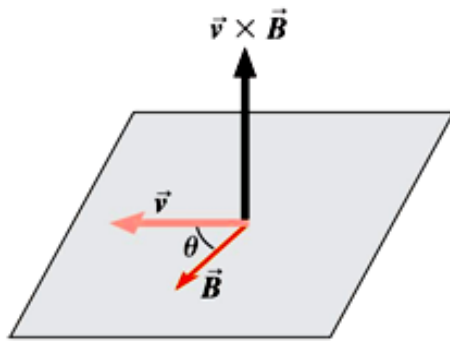
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- Magnetic Dipole

Magnetic force on moving charges: The Definition of **B**

$$\vec{F} = q\vec{E} \quad (\text{electric force on a charged particle})$$

$$\vec{F} = q\vec{v} \times \vec{B} \quad (\text{magnetic force on a moving charged particle})$$



$$F_B = q v B \sin \theta$$

$$[B] = \text{N} \cdot \text{s} / (\text{C} \cdot \text{m}) = \text{T (Tesla)} = 10,000 \text{ G (Gauss)}$$

$$B_{\text{Earth}} \sim 1 \text{ G}$$

$$B_{\text{small magnet}} \sim 100 \text{ G}$$

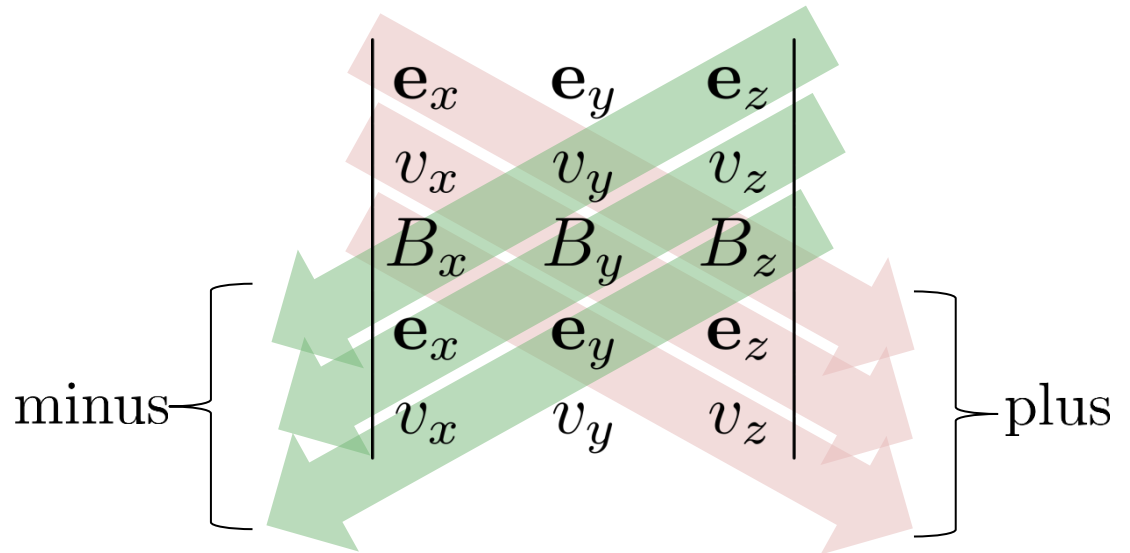
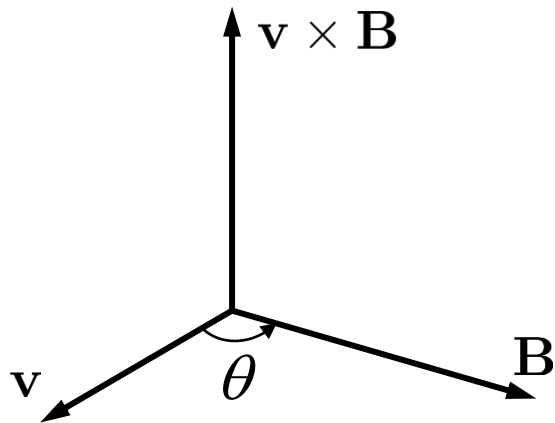
$$B_{\text{MRI}} \sim 1 \text{ T}$$

磁共振成像

A little math on magnetic force

To get the x, y, z components of a magnetic force, according to the definition of a cross product, we can use the following determinant form:

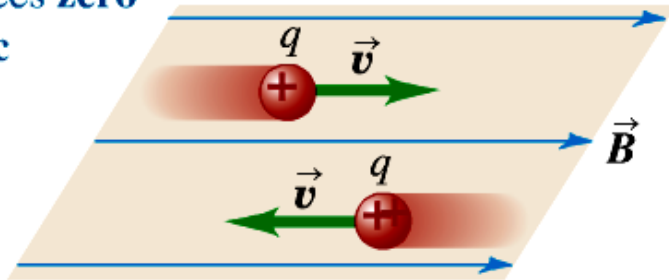
$$\begin{aligned}\mathbf{F} &= q\mathbf{v} \times \mathbf{B} = q \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix} \\ &= q[(v_y B_z \mathbf{e}_x + v_z B_x \mathbf{e}_y + v_x B_y \mathbf{e}_z) - (v_z B_y \mathbf{e}_x + v_x B_z \mathbf{e}_y + v_y B_x \mathbf{e}_z)] \\ &= q[(v_y B_z - v_z B_y) \mathbf{e}_x + (v_z B_x - v_x B_z) \mathbf{e}_y + (v_x B_y - v_y B_x) \mathbf{e}_z]\end{aligned}$$



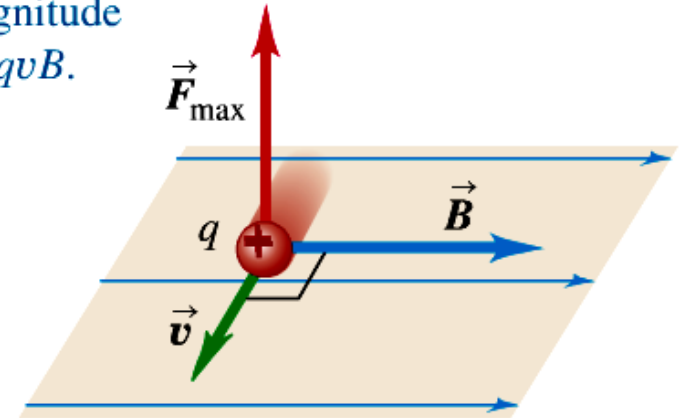
Magnetic force on moving charges

$$\vec{F} = q\vec{v} \times \vec{B} \text{ (magnetic force on a moving charged particle)}$$

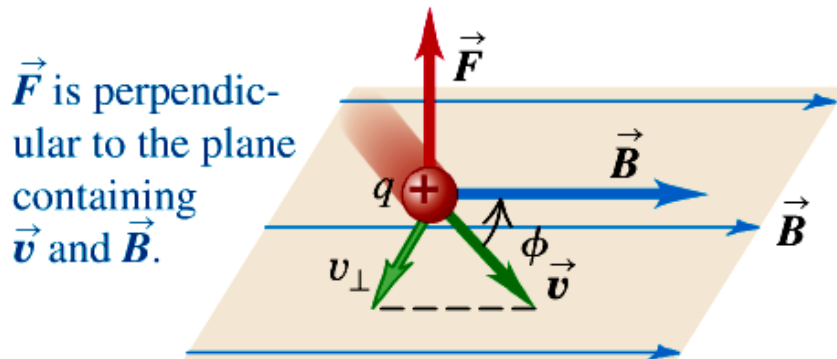
A charge moving **parallel** to a magnetic field experiences **zero magnetic force**.



A charge moving **perpendicular** to a magnetic field experiences a maximal magnetic force with magnitude $F_{\max} = qvB$.



A charge moving at an angle ϕ to a magnetic field experiences a magnetic force with magnitude $F = |q|v_{\perp}B = |q|vB \sin \phi$.



$\vec{F}$ is perpendicular to the plane containing $\vec{v}$ and $\vec{B}$.

$\therefore \vec{F} \perp \vec{v}$ **no work done by magnetic forces!**

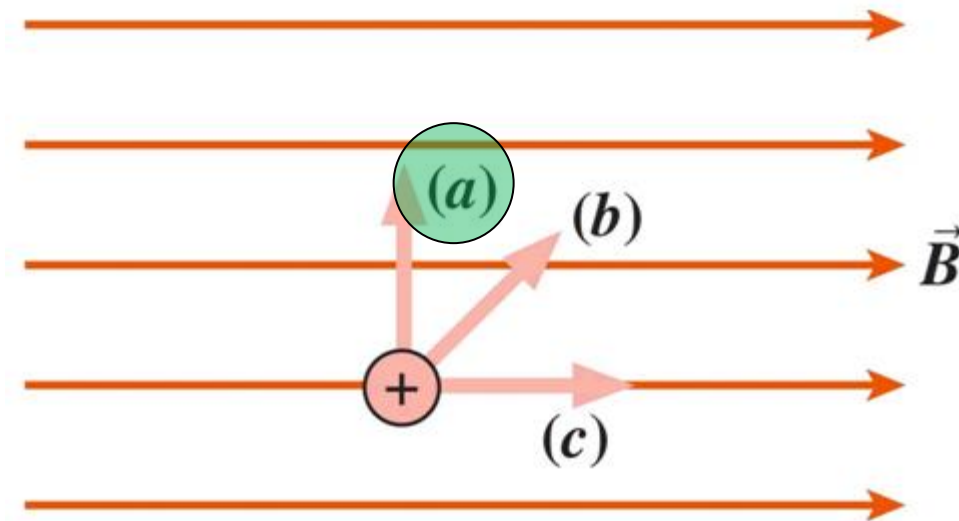
$\therefore$ speed v doesn't change
 $\rightarrow$ KE doesn't change!

Quick check

Figure shows a proton in a magnetic field.

For which of the three proton velocities shown will the magnetic force be greatest?

What will be the direction of the force in all three cases? $\perp$ & into paper



The Magnetic Force – Example

P747, Problem 56

A proton moves through a uniform magnetic field given by

$$\mathbf{B} = (10\mathbf{e}_x - 20\mathbf{e}_y + 25\mathbf{e}_z) \text{ mT}.$$

At time t_1 , the proton has a velocity given by

$$\mathbf{v} = v_x\mathbf{e}_x + v_y\mathbf{e}_y + (2.0 \text{ km/s})\mathbf{e}_z$$

and the magnetic force on the proton is

$$\mathbf{F}_B = (4.0 \times 10^{-17} \text{ N})\mathbf{e}_x + (2.0 \times 10^{-17} \text{ N})\mathbf{e}_y.$$

At that instant, what are v_x and v_y ?

The Magnetic Force – Example

Solve:

The solution to this problem is really straight-forward. We just use

$$\mathbf{F}_B = q\mathbf{v} \times \mathbf{B}$$

To find v_x and v_y .

$$\begin{aligned}\mathbf{F}_B = q\mathbf{v} \times \mathbf{B} &= q \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix} \\ &= q[(v_y B_z - v_z B_y)\mathbf{e}_x + (v_z B_x - v_x B_z)\mathbf{e}_y + (v_x B_y - v_y B_x)\mathbf{e}_z]\end{aligned}$$

The Magnetic Force – Example

i.e.,

$$\begin{cases} x \text{ component : } F_x = q(v_y B_z - v_z B_y) & \rightarrow & v_y = \frac{1}{B_z} \left(\frac{F_x}{q} + v_z B_y \right) \\ y \text{ component : } F_y = q(v_z B_x - v_x B_z) & \rightarrow & v_x = \frac{1}{B_z} \left(v_z B_x - \frac{F_y}{q} \right) \end{cases}$$

Plug in known numbers, we have

$$\begin{cases} v_y = \frac{1}{25 \text{ mT}} \left[\frac{4.0 \times 10^{-17} \text{ N}}{1.6 \times 10^{-19} \text{ C}} + (2.0 \text{ km/s})(-20 \text{ mT}) \right] = 8.4 \text{ km/s;} \\ v_x = \frac{1}{25 \text{ mT}} \left[(2.0 \text{ km/s})(10 \text{ mT}) - \frac{2.0 \times 10^{-17} \text{ N}}{1.6 \times 10^{-19} \text{ C}} \right] = -4.2 \text{ km/s.} \end{cases}$$

Moving charge in B-field

For $\mathbf{v} \perp \mathbf{B}$, $F = q v B$.  Circular motion

$$F = qvB = m \frac{v^2}{r}$$

$$r = \frac{m v}{q B}$$

Period

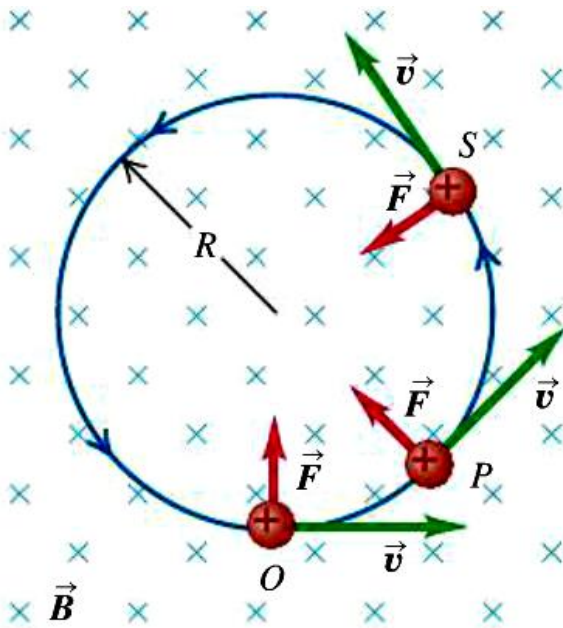
$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \frac{mv}{qB} = \frac{2\pi m}{qB}$$

Frequency

$$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

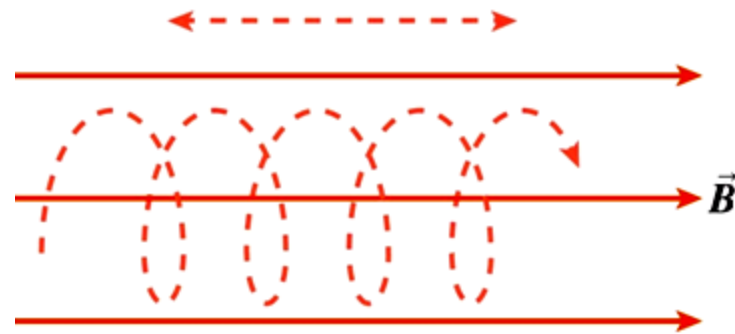
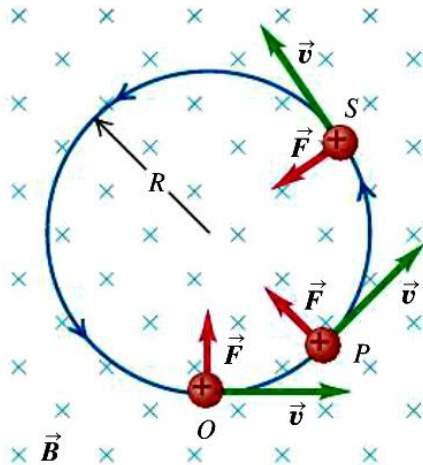
Angular frequency

$$\omega = 2\pi f = \frac{qB}{m}$$



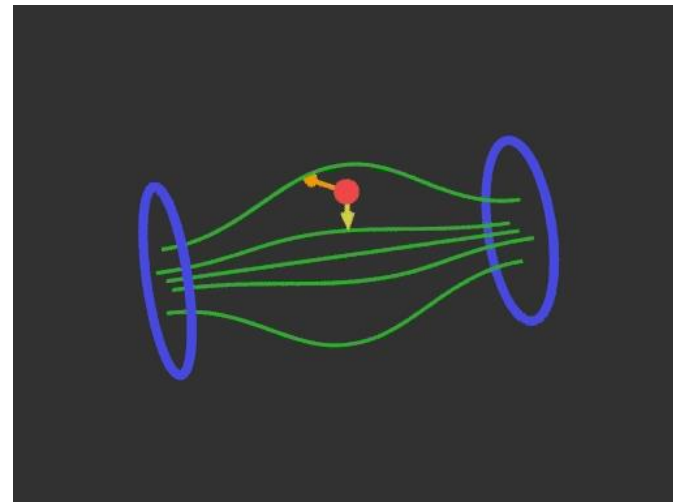
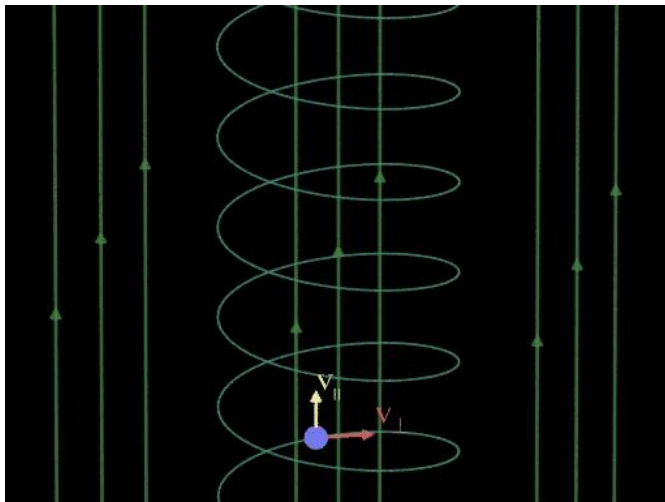
Helical motion

What if there is a v component parallel to the B field?



Motion // $\vec{B}$ not affected by it !

“Magnetic bottle”

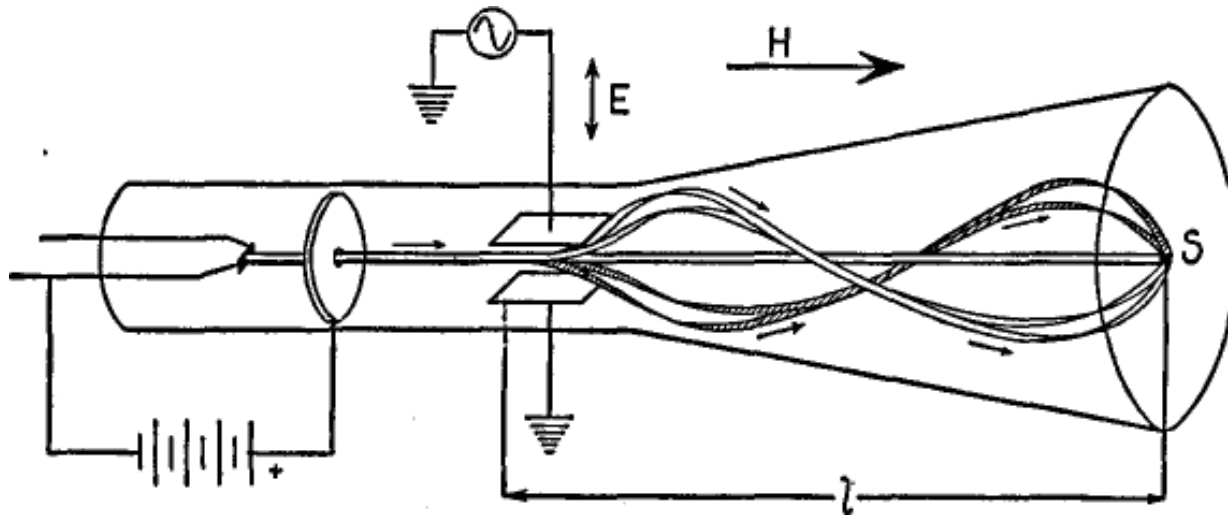


Magnetic Focusing

For charged particles propagate almost along $\mathbf{B}$,

$$p = \frac{2\pi m v \cos \phi}{|q|B} \approx \frac{2\pi m v}{|q|B}.$$

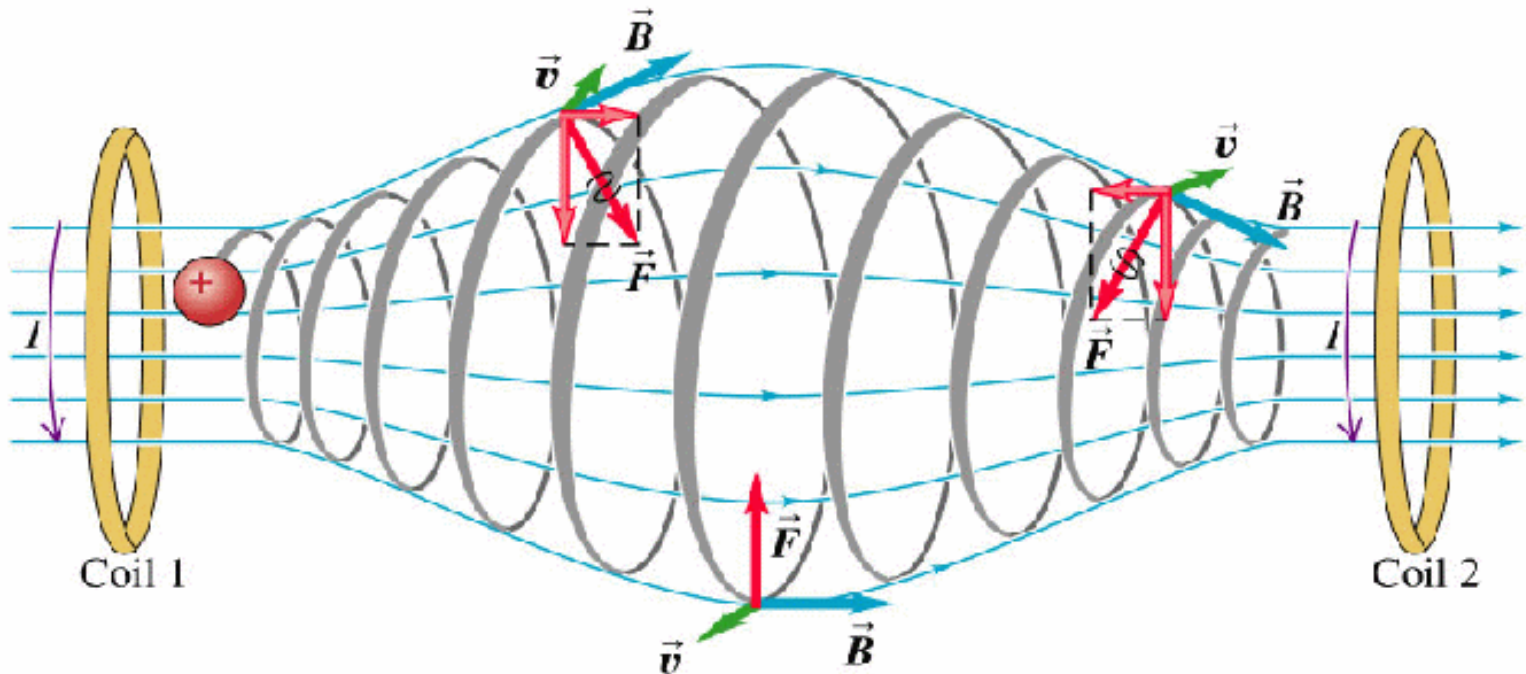
So the particles will be focused at $d = np$ (n is an integer).



Magnetic Focusing. From <http://www.vias.org/>

Magnetic Mirror

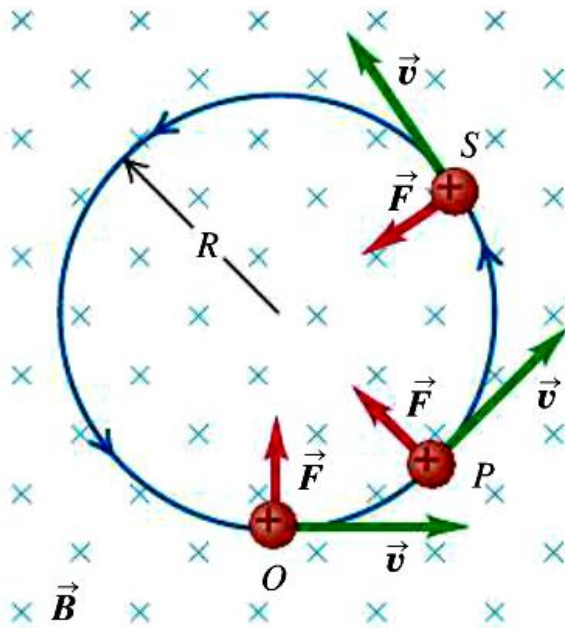
If gradient of field strength is introduced, $\vec{F}_B$ will have a component along the field direction to accelerate (decelerate) the particle. So the particles can be trapped in a certain space region.



Moving charge in B-field

For $\mathbf{v} \perp \mathbf{B}$, $F = q v B$.  Circular motion

$$F = qvB = m \frac{v^2}{r} \quad r = \frac{m v}{q B}$$



Period

$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \frac{mv}{qB} = \frac{2\pi m}{qB}$$

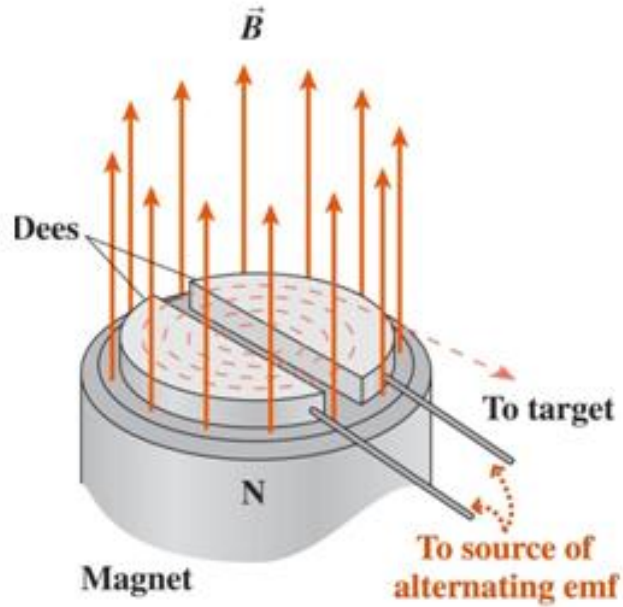
Frequency

$$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

Angular frequency

$$\omega = 2\pi f = \frac{qB}{m}$$

Cyclotron



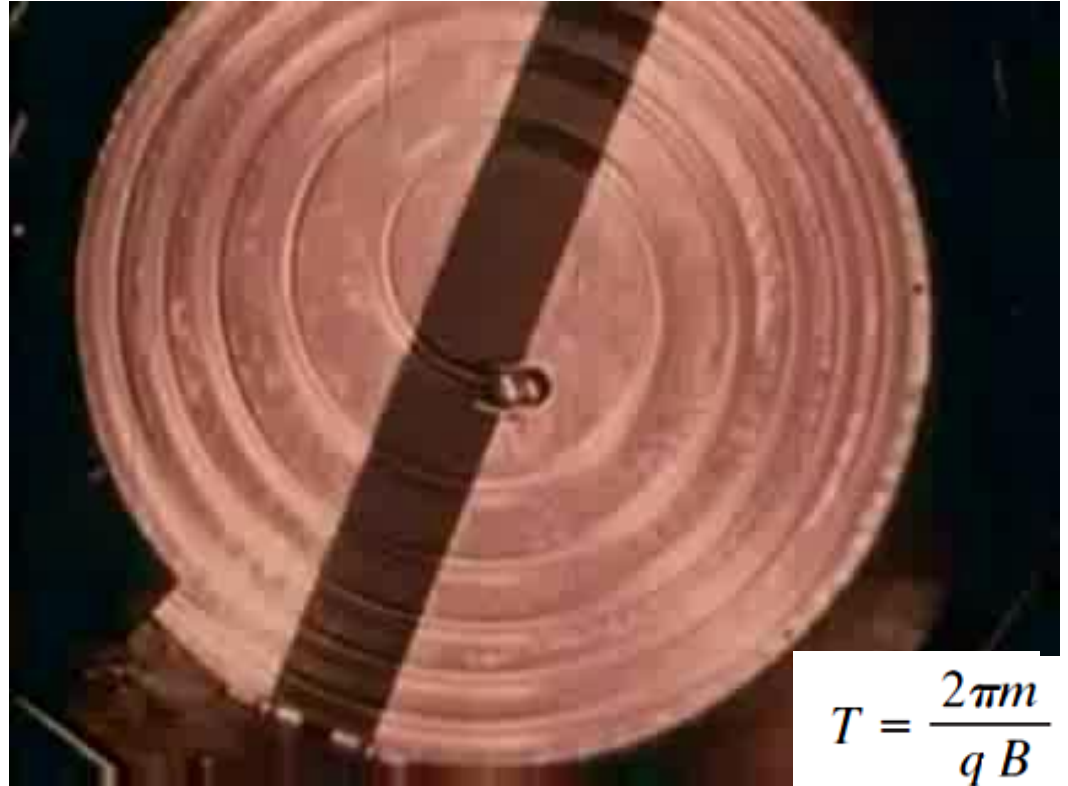
Whole device in vacuum chamber.

Small V across the dee's, which alternates polarities at the cyclotron frequency.

Particle injected at center of gap & spirals out.



Ernest O. Lawrence



$$T = \frac{2\pi m}{q B}$$

Cyclotrons and Synchrotrons

Synchrotron

A synchrotron uses time-varying magnetic field and oscillating frequency to overcome the relativity limit in the cyclotrons. The largest (most energetic) synchrotron in the world is the Large Hadron Collider (LHC) near Geneva, with a radius $r = 13.5$ km.



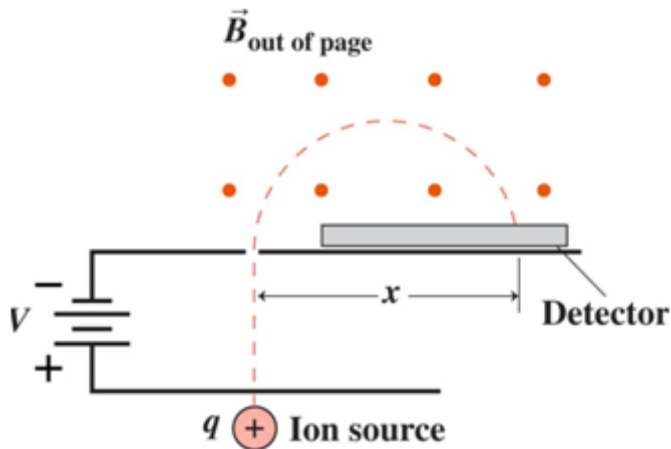
**A typical
synchrotron**
(Diamond Light
Source, Didcot, UK)

Mass Spectrometer

A mass spectrometer separates ions according to their ratio of charge to mass. Such devices are widely used to analyze unknown mixtures, and to separate isotopes of chemical elements.

Figure below shows ions of charge q & mass m being first accelerated from rest through a potential difference V , then entering a region of uniform $\mathbf{B}$ pointing out of the page.

Find an expression for the horizontal distance x .



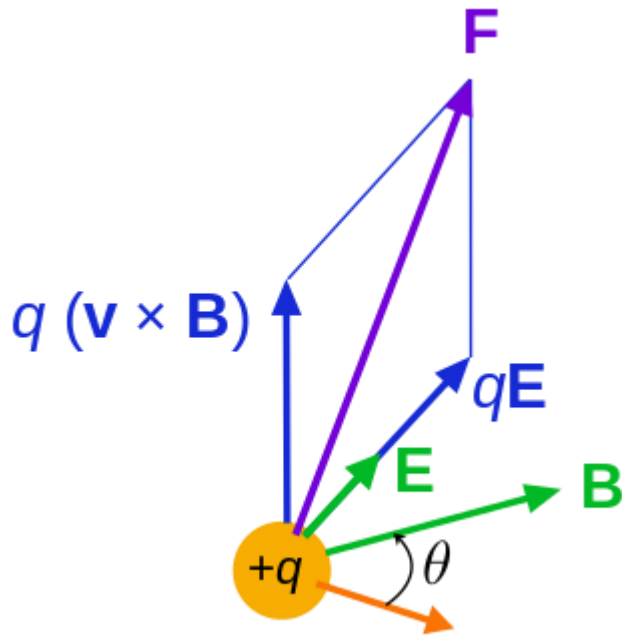
$$qV = \frac{1}{2}mv^2$$

$$m \frac{v^2}{r} = qvB$$

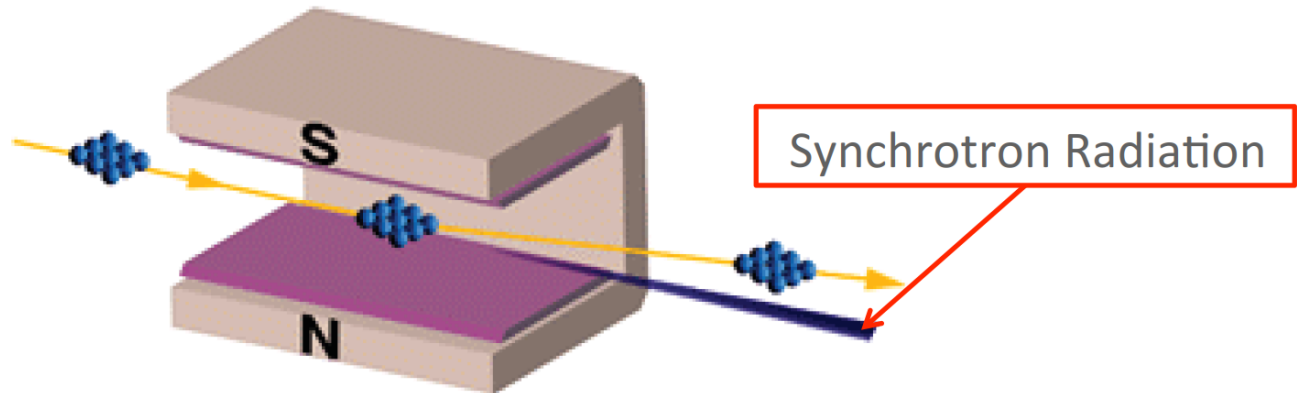
$$r = \frac{mv}{qB}$$

$$x = 2r = 2 \frac{mv}{qB} = 2 \frac{m}{qB} \sqrt{\frac{2qV}{m}} = \frac{2}{B} \sqrt{\frac{2mV}{q}}$$

Lorentz force: charge in electric and magnetic fields



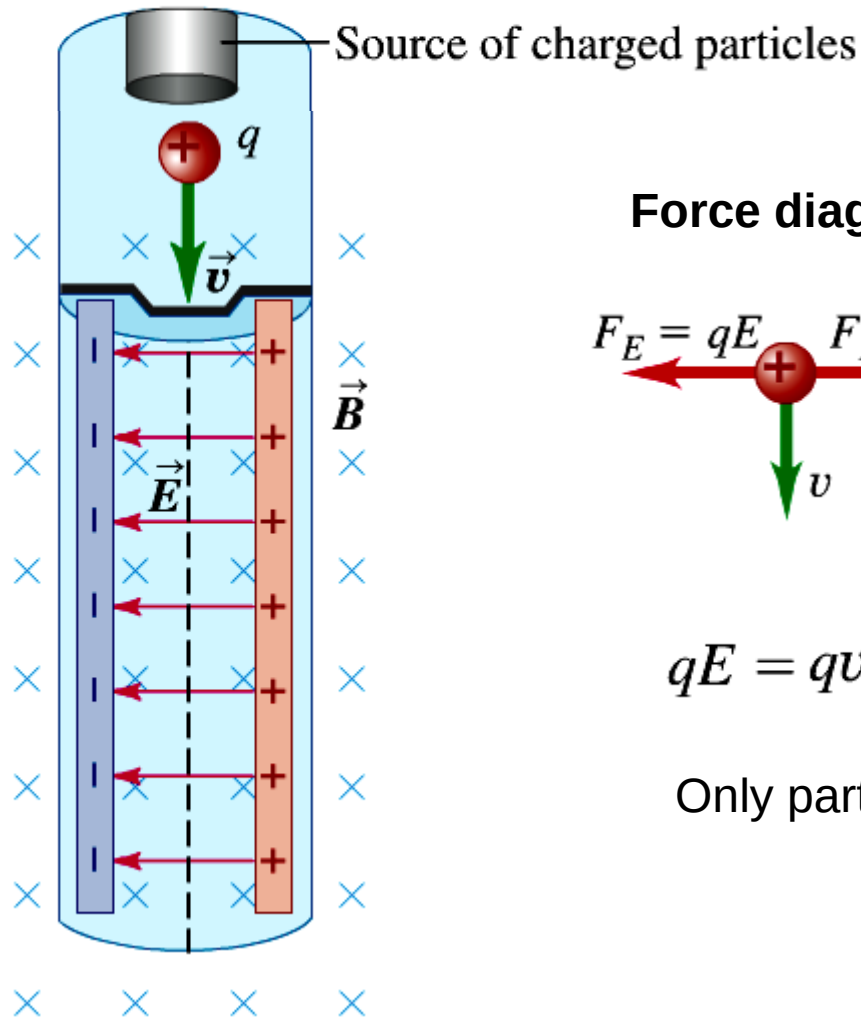
$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$



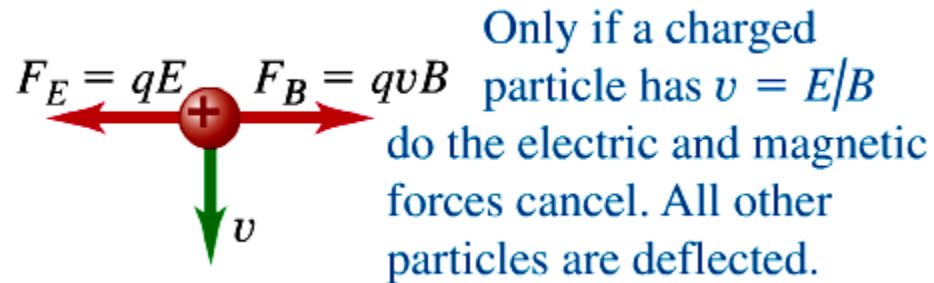
Chapter 28: magnetic force on moving charges

Chapter 29: magnetic field created by moving charges

Velocity selector



Force diagram

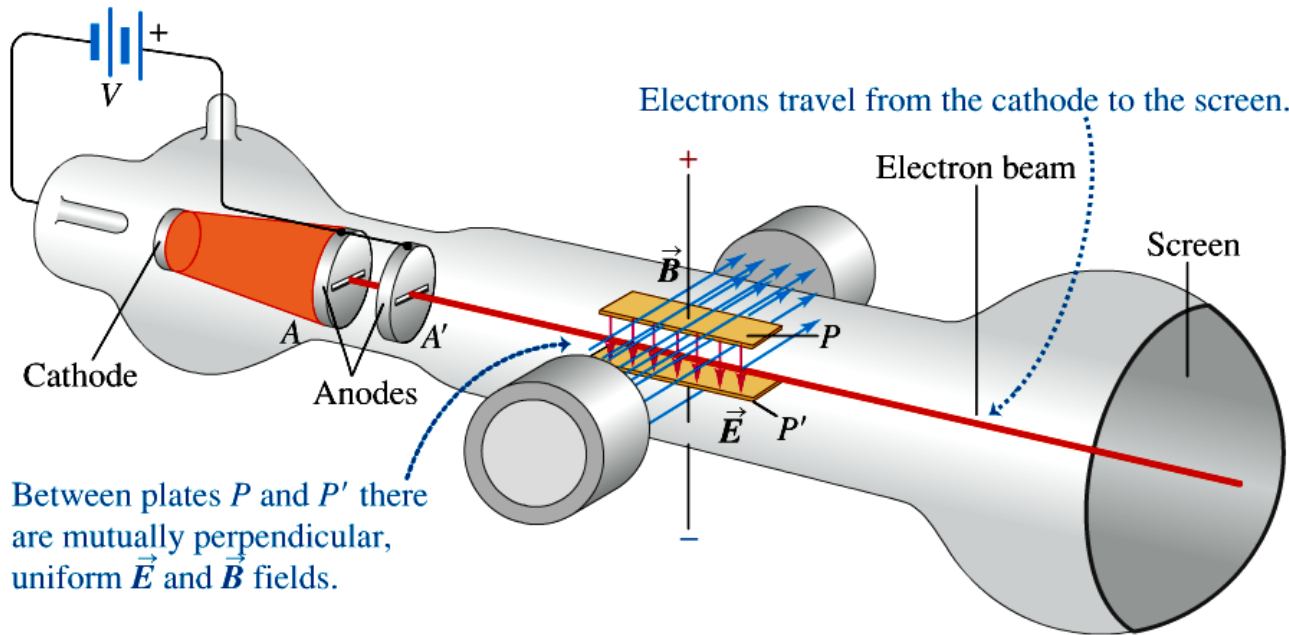


$$qE = qvB$$

Only particles with $v = \frac{E}{B}$ can pass through

Thomson's e/m Experiment in 1897

Discovery of the 1st subatomic particle: electron



Between A and A'

$$\frac{1}{2}mv^2 = eV$$

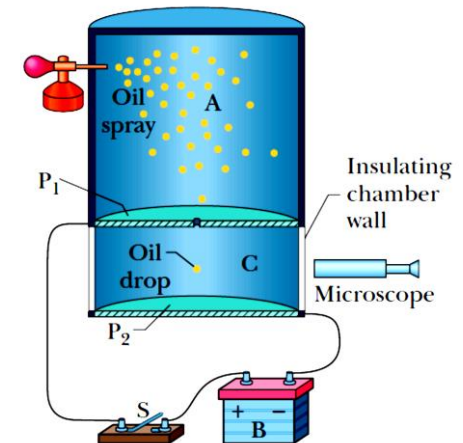
$$v = \sqrt{\frac{2eV}{m}}$$

$$\frac{E}{B} = \sqrt{\frac{2eV}{m}}$$

$$e/m = 1.758820150 \times 10^{11} \text{ C/kg}$$

$$\frac{e}{m} = \frac{E^2}{2VB^2}$$

$$m = 9.10938215(45) \times 10^{-31} \text{ kg}$$

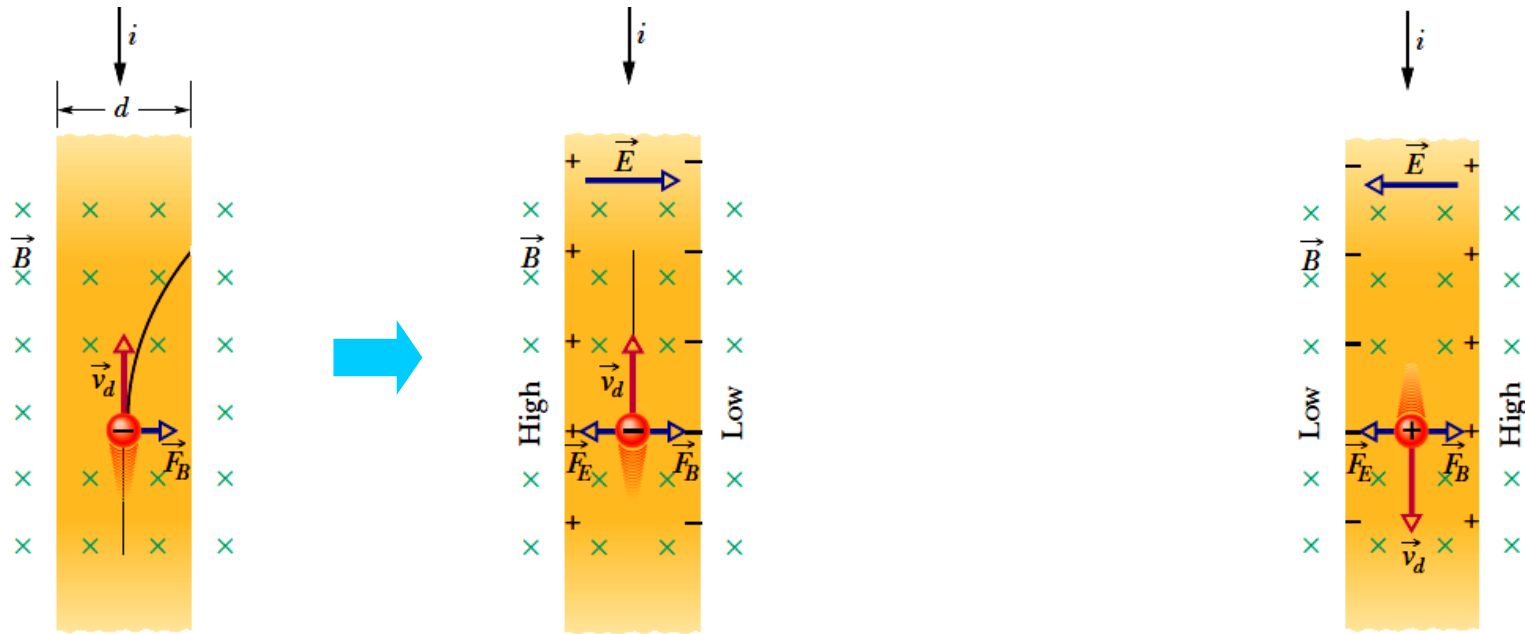


R. A. Millikan, *Phys. Rev.* 2, 109 (1913)

Today's topics (Chapter 28)

- Magnets
- Lorentz Force and Its Applications
- Hall Effect and Magnetic Force on a Current
- Magnetic Dipole

Hall effect



If the carrier is $-e$

If the carrier is $+e$

By a voltmeter, we can get V and $V = Ed$, $eE = ev_d B$.

From previous chapter

$$v_d = \frac{J}{ne} = \frac{i}{neA}$$

n : number of carrier per volume

$$n = \frac{Bi}{Vle}$$

l : thickness of the copper strip

Hall effect can be used to determine the type and density of charge carriers in a conductor.

More on Hall effect



Klaus von Klitzing



Robert B. Laughlin



Horst L. Störmer

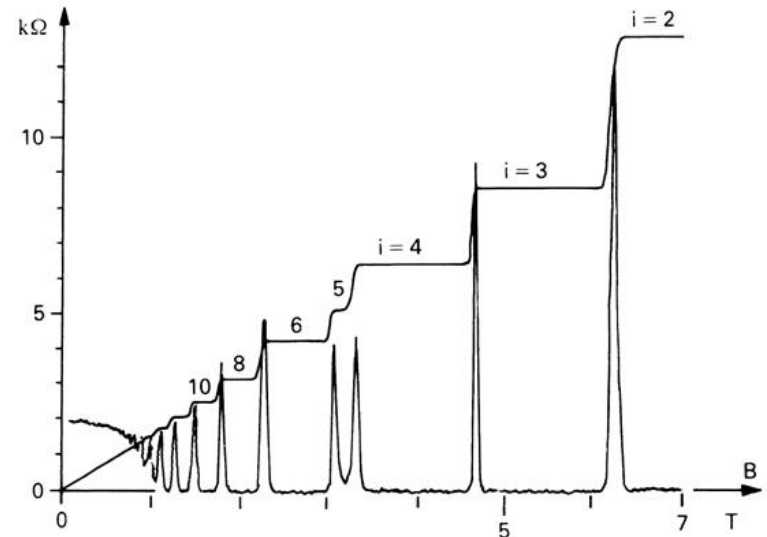


Daniel C. Tsui

Quantum Hall effect: Hall conductance σ takes on quantized values

$$\sigma_H = \nu \frac{e^2}{h}$$

where ν is an **integer** (Klitzing) or a **rational** number (Tsui).



More on Hall effect

极低温
强磁场

85 mK, 28 T

整数量子霍尔效应

$$\rho_{xy} = \frac{h}{ne^2}, \quad n = \frac{N}{N_\phi}$$

分数量子霍尔效应

$$n = 5/3, 4/3, 3/5 \dots$$

分数量子霍尔效应
获1998年诺贝尔物理奖



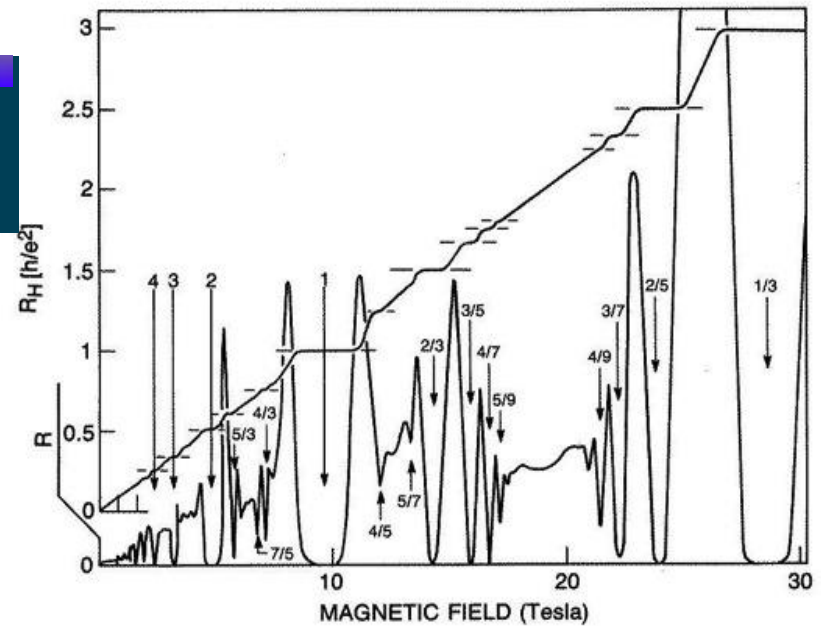
Robert B. Laughlin



Horst L. Störmer

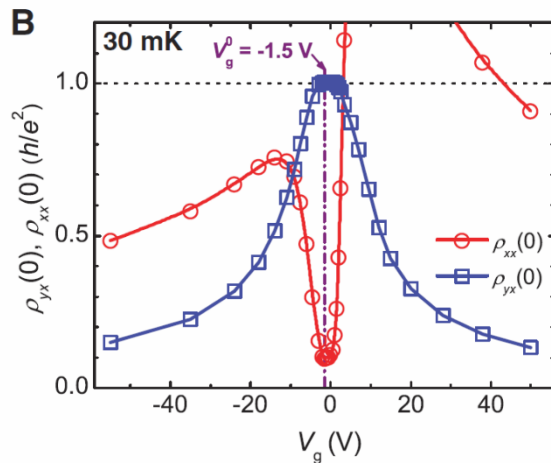
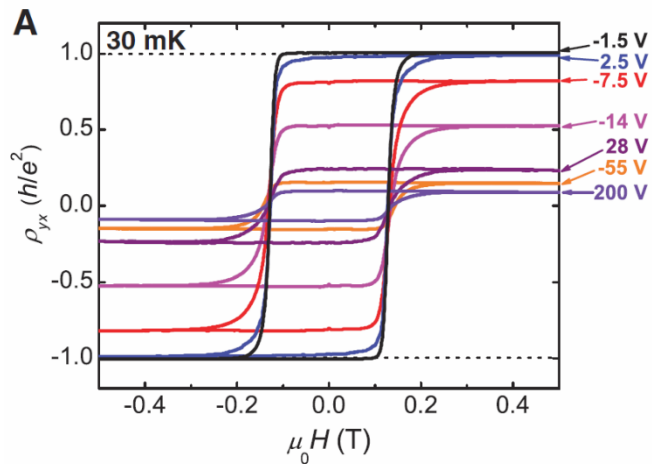


Daniel C. Tsui



More about the Hall effect

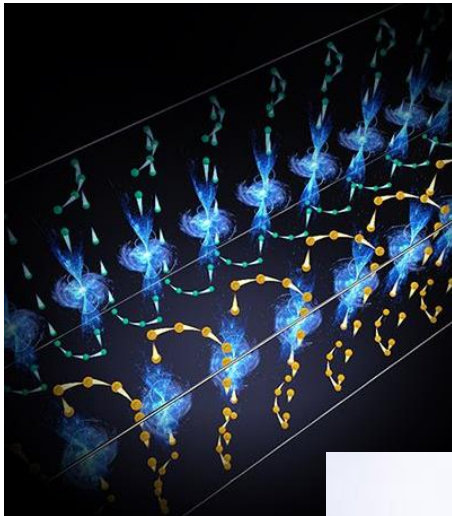
Quantum Anomalous Hall effect: A quantum Hall effect that can be observed even in zero field.



Qi-Kun Xue,
Tsinghua University,
China

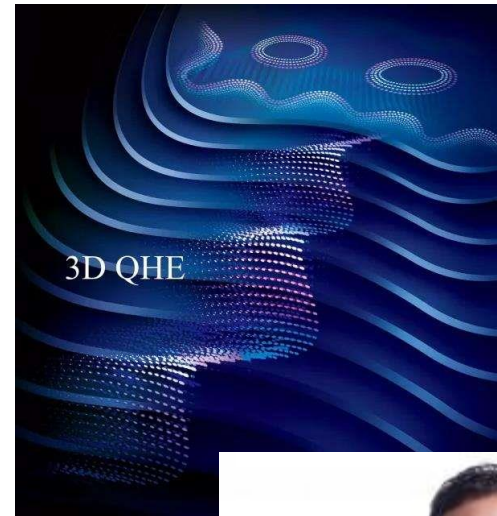
More about the Hall effect

Three Dimensional Quantum Hall effect: A quantum Hall effect that appear in thick samples but the conductance does not depend on the thickness.



The
“wormhole
tunneling”
version

Haizhou
Lu,
SUSTech,
China

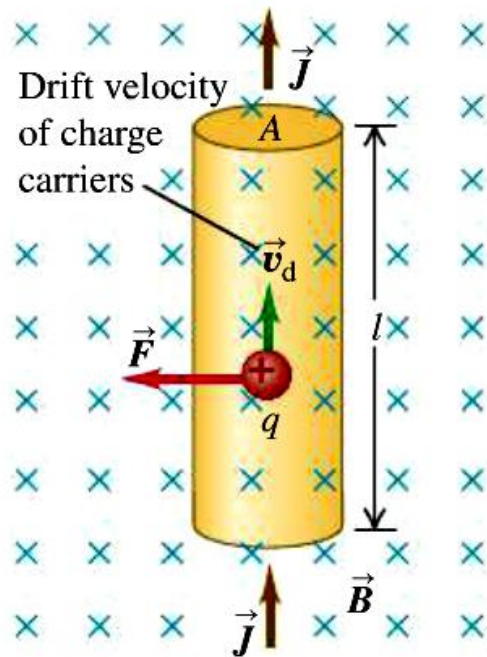


The “charge
density
wave”
version

Liyuan
Zhang,
SUSTech,
China



Magnetic force on a current carrying conductor



$$F = qv_d B \sin \phi$$

$$= \frac{i l}{v_d} v_d B \sin \phi$$

$$q = it = i \frac{l}{v_d}$$

$$\boxed{\vec{F} = i \vec{l} \times \vec{B}}$$

or we can start from :

$$F = (nAl)(qv_d B)$$

$$= (nqv_d A)(lB)$$

$$F = i l B$$

n is the number of charge per unit volume

From previous chapter :

$$J = nqv_d$$

$$JA = i$$

If l is not a straight line, we can write $d\vec{F} = i d\vec{l} \times \vec{B}$

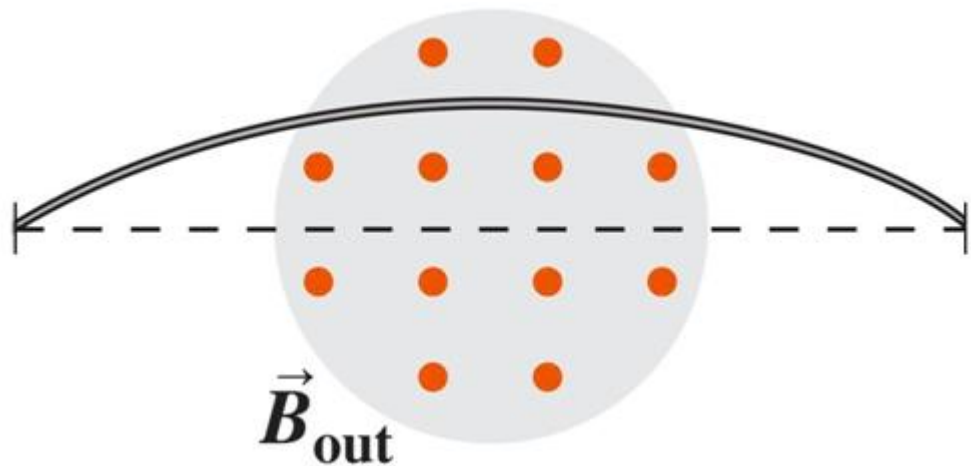
Example

Figure shows a flexible wire passing through a magnetic field that points out of screen.

The wire is deflected upward, as shown.

Is the current flowing

- (a) to the left or
- (b) to the right?



More Difficult Example

Wire with current i .
Magnetic field out of page.
What is net force on wire?

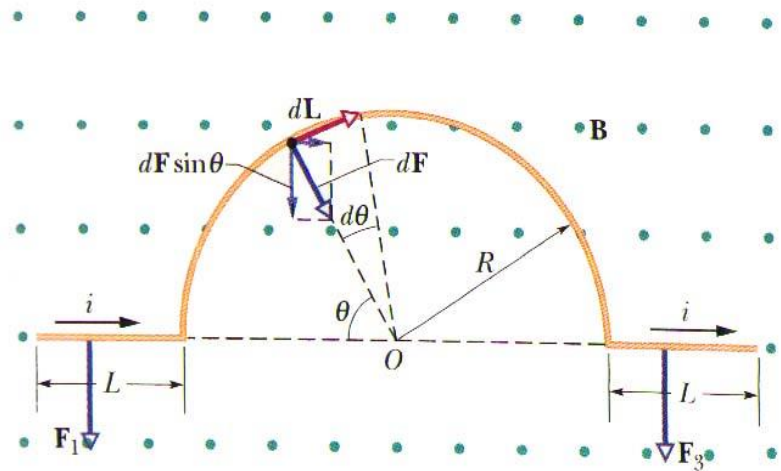
$$F_1 = F_3 = iLB$$

$$dF = iBdL = iBRd\theta$$

By symmetry, F_2 will only have a vertical component,

$$F_2 = \int_0^\pi \sin(\theta) dF = iBR \int_0^\pi \sin(\theta) d\theta = 2iBR$$

$$F_{\text{total}} = F_1 + F_2 + F_3 = iLB + 2iRB + iLB = 2iB(L + R)$$



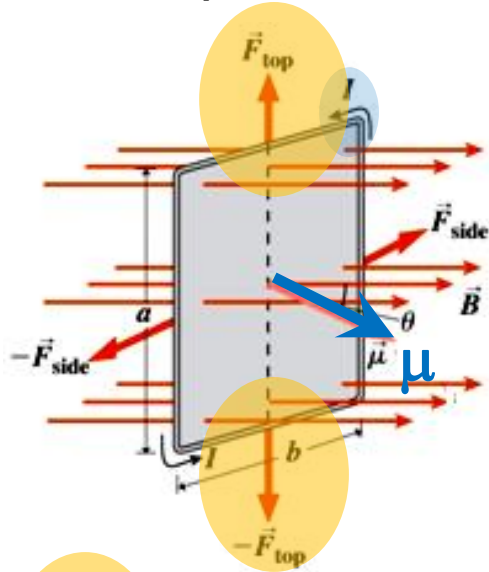
The net force acting on a curved current perpendicular to a uniform magnetic field doesn't depend on the shape

Today's topics (Chapter 28)

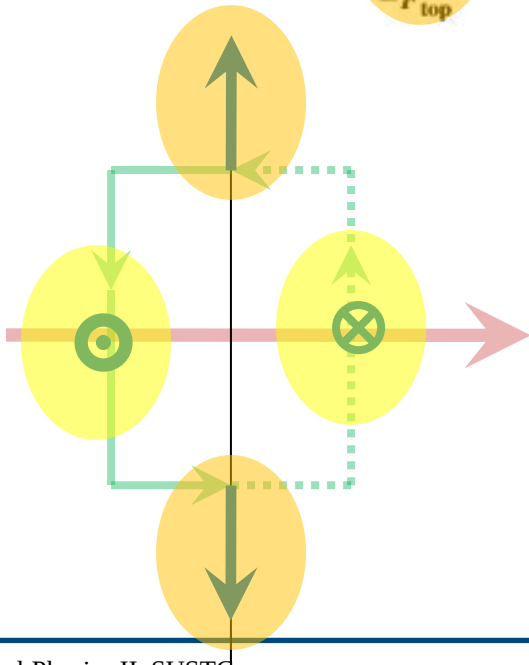
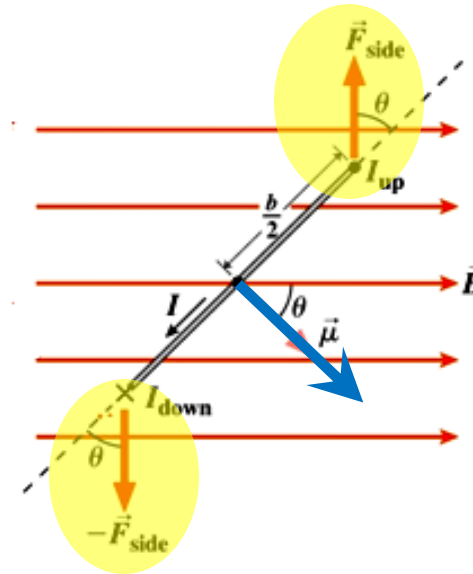
- Magnets
- Lorentz Force and Its Applications
- Hall Effect and Magnetic Force on a Current
- Magnetic Dipole

Torque on a Magnetic Dipole

Forces on top & bottom cancel



Forces on sides also cancel; but give net torque.



$$F_{side} = a I B$$

$$\tau_{side} = \frac{1}{2} b F_{side} \sin \theta$$

$$= \frac{1}{2} b a I B \sin \theta$$

$$\tau = 2\tau_{side} = b a I B \sin \theta$$

$$= \mu B \sin \theta$$

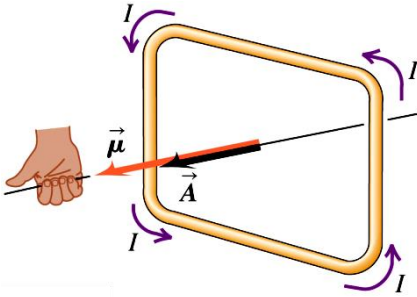
$$\mu = \text{area} \bullet \text{current}$$

$$\text{Torque on dipole: } \vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\text{Associated energy: } U = -\vec{\mu} \cdot \vec{B}$$

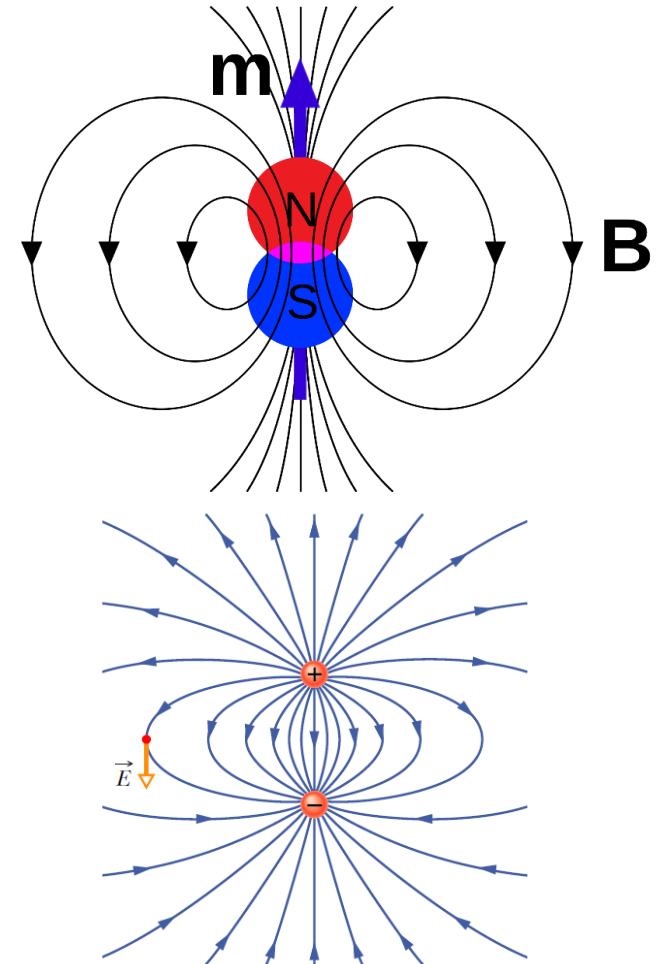
magnetic dipole moment
direction $\perp$ coil

Magnetic dipole and electric dipole



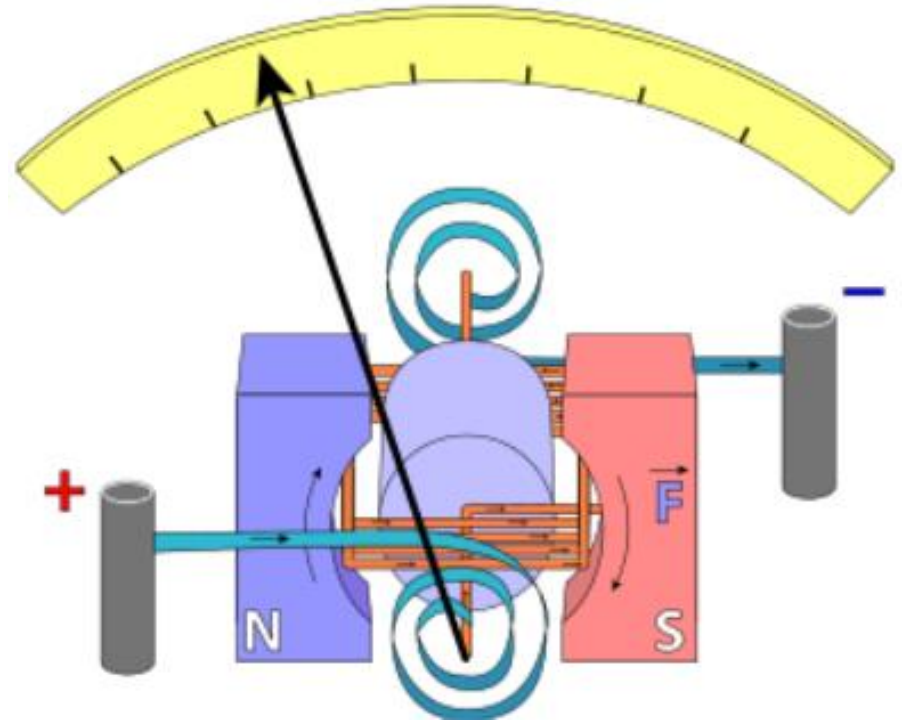
The right hand rule determines the direction of the magnetic moment

	Electric dipole	Magnetic dipole
Dipole moment	$\tau = \mathbf{p} \times \mathbf{E}$	$\tau = \boldsymbol{\mu} \times \mathbf{B}$
Potential energy	$U = -\mathbf{p} \cdot \mathbf{E}$	$U = -\boldsymbol{\mu} \cdot \mathbf{B}$



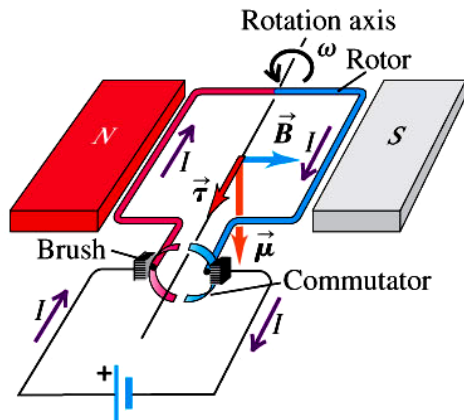
Application 1: Galvanometer (检流计)

Device used in the construction of ammeters and voltmeters.



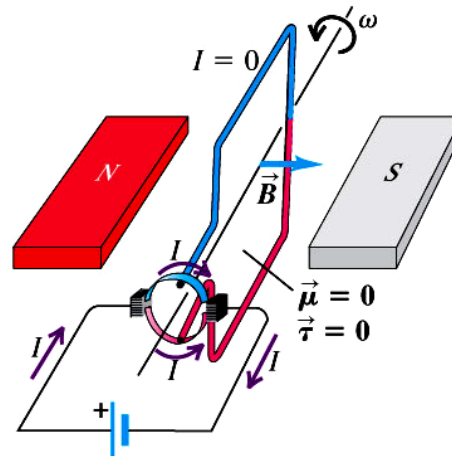
Application 2: DC motor

(a) Brushes are aligned with commutator segments.



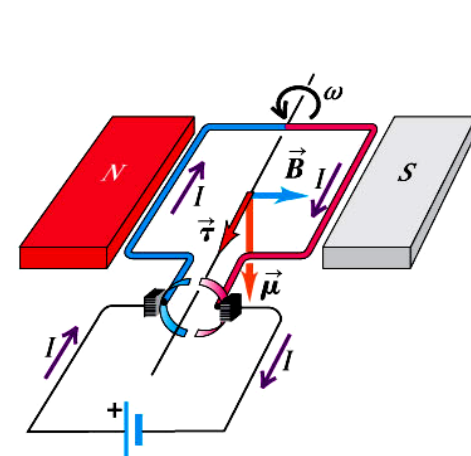
- Current flows into the red side of the rotor and out of the blue side.
- Therefore the magnetic torque causes the rotor to spin counterclockwise.

(b) Rotor has turned 90°.



- Each brush is in contact with both commutator segments, so the current bypasses the rotor altogether.
- No magnetic torque acts on the rotor.

(c) Rotor has turned 180°.

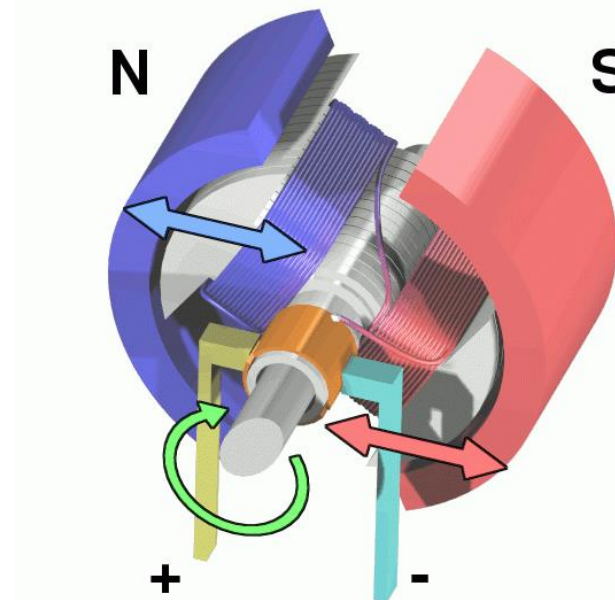
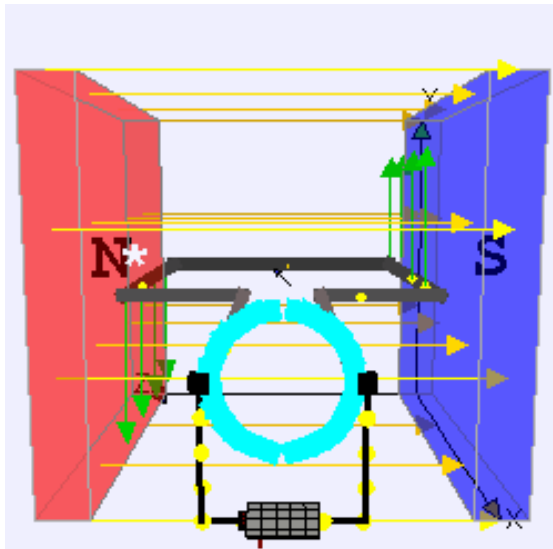


- The brushes are again aligned with commutator segments. This time the current flows into the blue side of the rotor and out of the red side.
- Therefore the magnetic torque again causes the rotor to spin counterclockwise.

Electric motors

This net torque causes the current loop to rotate inside the field, which has the following two very important applications:

1. **Electric motor:** the loop rotates continuously due to an ac-field.
2. **Power generator:** the loop is forced to rotate, generating ac-current.



Summary

Review & Summary

Magnetic Field $\vec{B}$ A magnetic field $\vec{B}$ is defined in terms of the force $\vec{F}_B$ acting on a test particle with charge q moving through the field with velocity $\vec{v}$:

$$\vec{F}_B = q\vec{v} \times \vec{B}. \quad (28-2)$$

The SI unit for $\vec{B}$ is the **tesla** (T): $1 \text{ T} = 1 \text{ N}/(\text{A} \cdot \text{m}) = 10^4 \text{ gauss}$.

The Hall Effect When a conducting strip carrying a current i is placed in a uniform magnetic field $\vec{B}$, some charge carriers (with charge e) build up on one side of the conductor, creating a potential difference V across the strip. The polarities of the sides indicate the sign of the charge carriers.

A Charged Particle Circulating in a Magnetic Field A charged particle with mass m and charge magnitude $|q|$ moving with velocity $\vec{v}$ perpendicular to a uniform magnetic field $\vec{B}$ will travel in a circle. Applying Newton's second law to the circular motion yields

$$|q|vB = \frac{mv^2}{r}, \quad (28-15)$$

from which we find the radius r of the circle to be

$$r = \frac{mv}{|q|B}. \quad (28-16)$$

The frequency of revolution f , the angular frequency ω , and the period of the motion T are given by

$$f = \frac{\omega}{2\pi} = \frac{1}{T} = \frac{|q|B}{2\pi m}. \quad (28-19, 28-18, 28-17)$$

Magnetic Force on a Current-Carrying Wire A straight wire carrying a current i in a uniform magnetic field experiences a sideways force

$$\vec{F}_B = i\vec{L} \times \vec{B}. \quad (28-26)$$

The force acting on a current element $i d\vec{L}$ in a magnetic field is

$$d\vec{F}_B = i d\vec{L} \times \vec{B}. \quad (28-28)$$

The direction of the length vector $\vec{L}$ or $d\vec{L}$ is that of the current i .

Torque on a Current-Carrying Coil A coil (of area A and N turns, carrying current i) in a uniform magnetic field $\vec{B}$ will experience a torque $\vec{\tau}$ given by

$$\vec{\tau} = \vec{\mu} \times \vec{B}. \quad (28-37)$$

Here $\vec{\mu}$ is the **magnetic dipole moment** of the coil, with magnitude $\mu = NiA$ and direction given by the right-hand rule.

Orientation Energy of a Magnetic Dipole The orientation energy of a magnetic dipole in a magnetic field is

$$U(\theta) = -\vec{\mu} \cdot \vec{B}. \quad (28-38)$$

If an external agent rotates a magnetic dipole from an initial orientation θ_i to some other orientation θ_f and the dipole is stationary both initially and finally, the work W_a done on the dipole by the agent is

$$W_a = \Delta U = U_f - U_i. \quad (28-39)$$